

"It's Something to Polish Your Own Thoughts, Rather than Create Thoughts for You": Understanding Participants' Use of Chatbots and LLMs During Online Research Participation

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Abstract

There are growing discussions within the research community about how to adapt study design given the widespread availability of Generative Artificial Intelligence (GenAI), including Large Language Models (LLMs). While much prior research has focused on LLM use from a researcher perspective (e.g. detecting and screening for LLM use) we present a complementary study from the perspective of participants who use LLMs during their research participation. In this exploratory interview study with 17 participants, we found a range of LLM use cases, from sourcing studies, to generating or modifying responses, to asking clarification questions about studies. We also explored participants' ethical considerations, finding that participants considered researchers' needs for authentic data when setting ethical boundaries. Participants also discussed how attempts to thwart their LLM use have negatively impacted their everyday participant experience. We propose a set of recommendations that researchers can incorporate into their studies to proactively address participant LLM use.

CCS Concepts

• Human-centered computing → User studies.

Keywords

participant experience, LLMs, GenAI, data quality, participant ethics, chatbots

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1 Introduction

Large Language Models (LLMs), algorithms trained to generate human-like text and perform complex language tasks, are becoming an increasingly common tool amongst the internet population. In a recent study among American college students, over a third of

respondents reported using an LLM for writing at least monthly [2]. These models are being used for a wide variety of tasks from cooking recipe generation [13] and home automation [6]. Researchers have found a variety of applications of LLMs to improve the quality and velocity of their own research, automating tedious and time-consuming tasks like qualitative coding [14] and writing research outlines [21].

Within the research community there is a growing discussion about how, without changes to study design and participant screening, LLMs may be used by research participants in ways that can impact the validity and quality of online research [9, 22], leading some researchers to wonder whether in the future it will become more time- and cost-efficient to move studies from online to in-person [9].

In light of these ongoing discussions, we add a new perspective and turn to participants themselves to better understand how they are adopting this new technology. Understanding participants' key use cases and underlying motivations for LLMs are critical when determining holistic study design strategies that address data quality needs while maintaining an acceptable participant experience.

We conducted an exploratory interview study with 17 participants who incorporate LLMs into their everyday research participation, focusing on the following key research questions:

- How do participants incorporate LLMs into their research participation process? What are their key use cases, and why are they using LLMs in this way?
- How has their experience participating in studies changed, in light of LLMs becoming more widely available, if at all?
- How do participants think about the acceptable use of LLMs in their research participation?

We extend the existing research by contributing a deeper understanding of the needs participants are fulfilling by using LLMs during their participation. While creating and editing text-based responses is a key use case, participants also use LLMs in ways that would be otherwise unknown to researchers, such as asking the LLM to clarify confusing study instructions while taking a survey. We also uncover the types of ethical frameworks participants adopt to guide what they consider to be acceptable LLM use.

2 Prior Work

Within the domain of survey research, researchers have always needed to consider the accuracy and quality of responses when designing questions, survey platforms, analysis procedures, and filtering criteria. Even a survey conducted on paper might be filled out in a rush without a respondent reading all of the questions.

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Successive waves of technological changes in how surveys are deployed have required new techniques to ensure the quality of survey responses.

2.1 Early approaches to ensure survey response quality

In the 1960s, researchers innovating on fielding surveys by mail discussed techniques to combat the practice of respondents "passing on" paper surveys for others to fill out on their behalf [17]. As surveys moved online, multiple new threats to data quality emerged. Both "speeders," participants who rush through a survey choosing answers without reading the questions [11], and "bots," with scripts that complete surveys instead of actual respondents [20], have led to a rise in junk responses not representative of the broader population. As a result, online survey best practices now include the incorporation of CAPTCHAs and new types of attention check questions [1, 5, 7].

Survey design has also evolved to include more open-ended questions, where participants can share a few sentences on a topic. These open-ended questions could then be used to detect speeders, or to capture more detailed stories from participants. Bentley et al. explored the suitability of different survey platforms for capturing this type of open-ended data [4]. Before LLMs, it would be difficult for a participant to fabricate a large block of text that was relevant to the question without taking the time to write an original response. However, now these types of questions open the opportunity for LLM use to quickly create a reasonable yet inauthentic response.

2.2 Detecting LLM Responses

Increased use of LLMs is leading to new techniques among researchers to detect and screen out problematic data (e.g. [18]). Xu et al. [24] explore several ways to detect bot participants, which might also be relevant to today's LLMs. These include looking at the time taken to complete a question, adding honeypot questions, and the use of CAPTCHAs or attention checks in the survey.

Another avenue of detection is in determining if a particular text was written by an LLM [19]. Many products and toolkits exist to do this today, and Wu et al. [23] provide a literature review of the different solutions currently available. It is important to note that these tools are not foolproof, and many have high error rates. They should be used with care, especially if used to disqualify participants from compensation. For example, Lebrun et al. [9] found an accuracy rate across seven AI detection systems of only 43%.

2.3 Preventing LLM Responses

Survey platforms are struggling to keep up with controls for preventing AI-generated responses. As LLMs typically run in another window with results pasted back into the surveys, there are hacks and discussions on how to disable pasting responses on survey platforms such as Qualtrics ¹.

Other research has worked to create new screening questions that are more visual and less prone to being answerable by an LLM. For example, Serafini et al. [18] created more visual screening

questions to determine if a respondent was a software developer. These types of questions may become less viable with newer multi-modal models.

2.4 Participant experience in online research

Other related work has explored the participant experience in online research. Schirra et al. [16] investigated the types of problems participants face when participating in online research studies. These include struggling to understand confusingly worded questions and unpaid disqualification from studies after significant investments in screening processes. In the present study, we found that frustration with screening processes was one reason that participants turned to LLMs during research studies.

New platforms have been created to completely remove the user from user research. Products such as Synthetic Users ² use AI models to complete UX research without involving human participants. Li [10] describes issues with these approaches and the lack of detailed responses and rich stories when compared to traditional research with people. We found a similar refrain amongst participants who valued their own experiences over those crafted by a computer.

3 Methods

We recruited participants through an online research platform to participate in a remote, unmoderated structured interview study. To qualify, participants needed to report using LLMs for any aspect of their research participation, participate in research studies at least a few times a month, have at least 6 months of research participation experience, and have an above beginner familiarity with LLMs. Qualified participants completed an informed consent process. Next, they began the structured interview, where they were presented with a series of open-ended questions and responded through the study platform's audio recording capabilities. Through these prompts, participants could describe their past experiences with LLMs in their own words.

Participant interviews were audio recorded and transcribed. They were asked about their experience participating in research, how they have been using LLMs as a part of their research practice, in what ways using LLMs has failed them in their research practice, and how they have seen study design change in response to the increased usage of LLMs. The resulting data was analyzed through a team-based, grounded-theory affinity diagramming exercise [3] (following Metcalf and Harboe [12]) where verbatim user quotes extracted from each transcript served as the base unit of analysis. Following best practices from these references, the quotes in the item level were the minimum length to capture a coherent thought from participants, typically 1-2 sentences from the interview transcript. For the first level of analysis, the authors collaboratively and inductively identified patterns across participant data, giving each group a descriptive label. Next, the authors grouped together multiple clusters and gave them explanatory labels, demonstrating broader patterns observed across participants. Finally, these pattern groups were organized into high-level categories that served as the structure for our study results, each grounded in data from multiple participants.

¹<https://community.qualtrics.com/custom-code-12/prevent-pasting-in-answers-23411>

²<https://www.syntheticusers.com/>

In total, 17 United States-based participants were accepted into the study. Participants were diverse with respect to age (22–46; mean=32), gender (8 women, 8 men, 1 non-binary participant), and level of self-reported expertise with LLMs (3 expert, 12 advanced, 3 intermediate). Interviews lasted about 10 minutes on average, and all participants received a \$10 gratuity for their time.

While all participants had at least 6 months of research participation experience, some had as many as 10 years. We purposefully recruited more experienced participants because we wanted them to draw from experiences across a variety of studies. Past research has shown that experienced research participants have a more in-depth understanding of the research process and platforms [15, 16].

4 Results

Participants reported a broad range of LLM-assisted and -generated activities during their research participation, primarily motivated by a desire to improve the quality and efficiency of their responses. However, this LLM use was not limited only to the study itself and also applied to sourcing research opportunities and completing research screening activities. While a few participants discussed recent examples of using LLMs to completely generate synthetic responses on their behalf, most felt LLMs only had a minor effect on their responses, and that their submissions were authentic to their experiences and opinions. Below we discuss these examples and motivations in more detail.

4.1 How LLMs are used by participants

Outside of the studies themselves, participants use LLMs in discovering new opportunities to participate in research studies. This includes activities such as understanding the types of research opportunities available on various platforms, how participants receive gratuities, and even if there are known user experience issues with the platform. As P16 describes, "I started to [use LLMs when I became] overwhelmed with all the platforms, how to best invest my time, where to focus and what to prioritize to maximize my earnings."

4.1.1 Qualifying for studies. Once participants find appealing research opportunities and platforms, they undertake various screening processes where study providers ensure participants are qualified based on the study criteria. Participants described using LLMs to help brush up on the study topic or complete the screening process itself.

Background research. In some studies, participants may be asked to speak about a specialized topic based on their expertise, and LLMs can quickly refresh the participant about topics or concepts relevant to the study. For example, P15 is sometimes invited to participate as an expert in life sciences in academic studies. She uses LLMs to "quickly summarize research topics, and also explain complex terminology, or provide an overview of the research area" and "quickly grasp background information or concepts I may not fully understand... even though I am an expert in the field... I just need a little bit of productivity boost."

Completing screening processes. Many online platforms providing on-demand participants include a screening step that they must pass before qualifying to participate in the study. Participants

often only qualify for an incentive if they complete the screening process. In discussing their frustrations, they mentioned "wast[ing] a bunch of time doing all these different screeners" (P9) and turning to LLMs in times of frustration. "I do fill out quite a few screeners, and sometimes the screeners ask a question I simply ... [don't] know what in the world they're talking about. And the LLM will answer it for me."

P6 attempted to use an LLM during a live phone screening process with a research recruiter to qualify for a study: "Because it was live, I couldn't get the [LLM to answer] prompts fast enough. ... and for the sake of not trying to have extreme pauses over the phone, I ended up being like, I can't rely on this any longer. ... And then I ended up guessing [the screener question]... and they were like, yeah, we can't continue with this research participation because you clearly aren't a user of this product."

4.1.2 Participating in studies. Participants discussed multiple uses for LLMs during their research participation. While the main use cases involved generating and modifying text responses, there were also examples of using LLMs to better understand study instructions and provide feedback on responses.

Clarifying study instructions. As noted in past research [16], participants may experience confusing study instructions, particularly when participating through on-demand research platforms that lack live assistance from the researcher. LLMs can serve as a tool for deciphering and breaking down confusing instructions. As P15 notes, "When a task or instructions feel unclear, I consult LLMs to break down complex information into simpler terms."

Real-time response feedback. Most unmoderated study platforms lack robust validation tools that help participants understand if their responses follow all of the instructions, particularly as they relate to the content of responses. Meeting all requirements is critical, as responses that do not meet the guidelines may be rejected later. Participants reported using LLMs both to validate that their response meets the technical requirements—" [LLMs help] me stick to a set of rules, meaning the word count" (P2)—and also the qualitative requirements of certain open-ended responses. P8 noted, "It helps me do a better job in framing my response and making sure I incorporate all aspects to the question."

Creating hybrid responses. The largest category of use cases among participants related to LLM-assisted writing processes. In these cases, participants supplied the underlying ideas (whether through fragments or wholly written responses) and used the LLM to modify the text.

Participants described two key motivations for using LLMs for writing-related tasks. First, they wanted to increase their efficiency across studies. For example, P3 said, "It helps me write comprehensive and coherent responses in a shorter amount of time." Similarly, P11 noted, "I started using LLMs to just save myself some time, because, why not use LLMs? I would guide it on how to write an effective response, [and] I feel like it saves me a lot of time."

Response quality is also a consideration for LLM use. Participants described improving the quality of their responses as a kind of value-add for the researchers who will ultimately have to read their responses: "You basically say, Hey, LLM, these are the things I wanna say. And then have the LLM write... a really eloquent response..."

so someone [the researcher] is not trying to dig through, you know, your brain dump" (P8).

We call these *hybrid responses* because embedded in participants' descriptions of these behaviors is a sense that they are producing the original writing for the response, and the LLM is simply improving their contributions. Some of the hybrid writing tasks described by participants include:

- **Expanding or "bulking up" responses:** "They expect paragraphs and paragraphs of responses. So what I'll do is write my response out... and [the LLM] will suggest a more kind of polished, professional version" (P10).
- **Organizing or framing responses:** "If I'm asked a question, I'm like, I know the answer, and I start either verbalizing or writing something up. And then I'll realize I'm not really doing a great job at this, so I'll use an LLM to help me frame my response better" (P8).
- **Rewriting responses for clarity:** "Whenever I'm supposed to type some of my answers, because I'm in a rush, I don't want to spend so much time with editing, I might use it to help me get my answer a lot more clearer, a lot more concise" (P1).
- **Fixing grammar:** "As a non-native English speaker, [LLMs] help me avoid a lot of mistakes" (P11).

Creating synthetic responses. Participants also described cases where their contributions begin with an LLM-generated response that is often later edited: "I always edit and fine-tune what the LLM produces, but I use it as a general starting point" (P2). However, participants also described relying solely on LLM-generated responses at times when study tasks felt tedious and frustrating. As P5 describes:

I've been asked to write "love letters" or "breakup letters" to specific [computer] applications or programs, which I think is kind of silly because [a participant] is going to have to, A, write about it, and B, have to think about that. I have to give an inanimate object people-like qualities [to complete a study], which is really weird for me... I started using LLMs this way because I was presented with a question like that, and I was really frustrated at the time... I asked [the LLM] to write me a letter and it got it right just fine.

This type of completely synthetic response generation represented a small set of use cases described by participants, but does suggest a connection between participant experience and participants' decision to use LLMs.

4.2 Participants' ethical considerations

Despite the few examples of synthetic research responses, nearly all participants discussed ethical boundaries that guided their use of LLMs during research participation. One large theme in the data involved participants describing the need for "authenticity". As P10 summarized, "I don't think you should use [LLMs] to research [study] information because I feel like that's not authentic. You're not giving your authentic feelings," adding, "It's more something to polish your own thoughts, rather than create thoughts for you."

Participants generally understood that researchers were recruiting participants to understand genuine human experiences, and providing LLM-generated responses defeats that purpose. "What's the point of using [research panels] if you're not going to get good research out of it anyways?... I think research is about getting actual, honest human opinions, and a large language model is simply not going to provide that" (P13).

Rather than generating responses, most participants discussed response improvement as an ethical use case. In such instances, opinions and experiences provide the ground truth for the response, with guardrails if the response feels too robotic or inauthentic. "It's more like creating a hybrid text from what I originally wrote to what [the LLM] suggests, and then polishing it out so that it sounds professional, but still human... if it fails, if the text doesn't sound the way I want it to, I'll just use my original text" (P10).

4.3 Addressing LLM use in study design

Participants described how the studies they take are evolving to respond to increased LLM use among participants. One strategy is for researchers to have a dedicated acceptable use statement at the beginning of research studies to explicitly describe what type of LLM use is allowed, if any: "I've seen a lot more comments and disclaimers on the studies that are like, please do not use AI to generate responses for the writing portions of this" (P16).

Beyond acceptable use statements, participants have also seen an increase in descriptions of consequences for LLM use. "I do see that researchers are starting to threaten a bit more, 'Don't use these [LLM] products,' and explicitly say so, and if you use them you could be in violation" (P17). Penalties can be applied "if there's even a suspicion [of LLM use]" and "your research participation can be canceled, you won't get paid, your dataset will get rejected" (P2).

Participants have also experienced a variety of indirect study and platform design choices meant to thwart LLM use. One category relates to study questions that LLMs may struggle to answer. "I've seen questions that are designed to try to theoretically trick these bots and have them not give a valid response so they can be rejected" (P5). A second category is for studies to simply remove the ability for participants to copy and paste, which is otherwise the most efficient way of bringing LLM-generated or -assisted text from an external LLM onto the study platform.

Multiple participants experienced negative consequences from these increasingly used strategies. P2 had a dataset removed from a study after being suspected of using AI, but says he hadn't used AI during that study. Meanwhile, P8 needed to copy a web URL provided in the study instructions to a new browser window, but was blocked from copying text. P2 experienced a similar situation in which he was asked to paste his LinkedIn profile into a text box for a study: "[Then] I had to manually type it out."

Strategies like blocking copy/paste add friction, but do not prevent LLM use among respondents, as P11 notes: "It'll say 'no paste allowed,' which still means that I can generate text and I can put it in a different window and just retype it."

5 Discussion and implications for research design and practice

Beyond creating or modifying text-based answers, this research shows that participants are integrating LLMs into the full cycle of their research participation, from finding studies, to qualifying for research, to asking clarification questions about the study, to seeking feedback on study responses. In many cases, LLMs are used in ways that would be difficult to detect or understand by simply looking at raw study data.

What becomes clear when looking across participant experiences is that there are currently few norms or standards across studies and platforms regarding LLM use for participants. In many cases, participants themselves are left to decide what is or is not allowed, and many have created their own ethical frameworks. While some uses of LLMs may potentially benefit the researcher (e.g. clarifying participant language), others may materially impact the study itself (e.g. brainstorming approaches on how to best respond to a question). Many platforms do not yet have guidelines around acceptable use of LLMs, so this responsibility falls to the researcher.

Participants' wide range of hybrid response use cases suggests that these types of LLM-assisted responses may become an increasingly common practice. Working with LLMs solves a variety of problems for participants, particularly around efficiency and study protocol compliance. Participants expressed frustration with the difficulty of crafting longform textual responses, and found that writing alongside an LLM makes the process easier. While most participants described the need for authenticity in their responses, and took steps to ensure their hybrid responses genuinely reflected their opinions, we saw examples of participants creating responses through an LLM. For example, some participants described using LLMs to brainstorm an original idea, which they then edited to reflect their personal tone. Others have used LLMs to pass screeners they otherwise would have failed in order to access study opportunities.

Researchers can proactively address some of the reasons that participants turn to LLMs to begin with. Ensuring instructions and study tasks are written clearly, screening for language proficiency (if required), and explaining the level of polish and editing expected in the instructions for writing-based tasks may help participants complete studies with less pressure to turn to LLMs for assistance.

We observed how attempts to hinder LLM use may become frustrating or burdensome to participants simply trying to follow study protocols. For example, completely removing the ability to copy or paste text (at the study or platform level) may require participants to manually input legitimate study data, or make it difficult to access required supplemental study materials like external links. It should be the responsibility of the researcher to understand the policies and feature sets of the platforms they are using to conduct research, and ensure their study design is compatible.

Penalizing suspected LLM users in post-hoc data cleaning processes can also be problematic. Participants' legitimate responses may be incorrectly flagged as LLM generated, causing them to lose gratuities or even access to research platforms if their responses are rejected.

We strongly recommend that researchers adopt LLM acceptable use statements that are provided to participants as part of the study onboarding process. This statement should cover aspects including:

- To what extent participants may incorporate LLMs into their research process. This could include modifying existing answers for clarity and grammar, or using the LLM to explain protocol questions to them. Or a clear statement that LLMs may not be used at all.
- Which steps in a participant's response writing process LLMs may be used for. For example, LLMs can be used in a final editing step, but not for baseline idea generation or inspiration.
- What participants can expect to happen if LLMs are used outside of acceptable use. Consequences like notifying the platform or giving them a participant a poor rating may reduce unwanted behavior while setting expectations.

6 Limitations

This study represents a foundational exploration of LLM integration into online research workflows, focusing on identifying use cases rather than quantifying their prevalence. These use cases were reflective of our sample, which consisted of experienced research participants. As such, this study may not represent the perspectives and practices of novice or infrequent participants. Their motivations and use cases might differ significantly, particularly concerning efficiency, which was one key driver for experienced participants in this study.

Furthermore, the study may not fully capture the views of participants who use LLMs in abusive or antagonistic ways. Social desirability bias [8] may have led to the underreporting or omission of such perspectives, potentially obscuring important use cases that researchers should be aware of.

7 Conclusion

This research provides valuable early insights into how experienced participants leverage LLMs in their online research workflows. Future research should explore the prevalence of identified use cases on a larger scale and track their evolution over time, given the rapidly changing technological landscape.

Investigating the experiences and motivations of novice and infrequent research participants represents another crucial avenue for future work. This will help to paint a more complete picture of LLM integration into participant online research workflows and provide researchers with a broader understanding of its implications.

While researchers should remain open to the potential benefits of these technologies, establishing clear and explicit guidelines about acceptable and unacceptable LLM usage is crucial. Additional research is needed to understand how participants perceive and react to such guidelines. Furthermore, researchers must carefully consider the specific policies and limitations of each online research platform regarding LLM use before designing their research protocols. This awareness will help ensure the ethical and responsible integration of these tools into online research practices.

Finally, future studies should explicitly address the potential misuse of LLMs in research and actively seek to incorporate the

perspectives of those who might engage in such practices, to better understand and mitigate the associated risks.

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